Nonverbal Behavior Does Not Add Out-of-Session Predictive Power for Hand Strength in Broadcast Poker: A Preregistered Replication Failure and a Case Study in Identity-Contamination Artifacts

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Abstract

Televised poker commentary and a recent broadcast "AI bluff detector" popularized the claim that nonverbal behavior reveals hand strength on camera. We test that claim with a fully automated pipeline over 22 hours of publicly broadcast cash-game footage with exposed hole cards: game state is read from the broadcast graphics by OCR, hands are reassembled into decision windows, hand strength is labeled by Monte Carlo equity, and face and body features are extracted for camera-covered decisions of tracked players. Our central methodological finding is an identity-contamination artifact: under progressively stricter face-attribution hygiene, an apparent behavioral effect first strengthened to a nominally resolvable ΔAUC of $+0.096$ (95% CI $[+0.005, +0.181]$) and then vanished entirely ($+0.009$, $[-0.058, +0.074]$) once every enrolled reference face was verified. The spurious effect was produced by frames in which a lookalike neighbor was silently credited to the tracked player. A preregistered replication on two additional sessions, with the analysis frozen before download, decisively failed: pooled across four sessions, face and pose features change out-of-session AUC by -0.075 (95% CI $[-0.134, -0.020]$, $n = 292$ covered decisions), with approximately 91% power to detect the registered $+0.096$. Sensitivity analyses show the degradation is not an artifact of estimator variance: no regularization scheme rescues the features, a behavior-only model is systematically anti-predictive out of session (AUC 0.349), the effect persists within a single player across his four sessions, and only part of the inversion is explained by session-level camera-geometry offsets. We conclude that at broadcast scale and with tabular behavioral features, nonverbal behavior adds no out-of-session predictive power over betting action, and that identity hygiene, not feature engineering, is the binding methodological risk for behavioral video datasets built on multi-person footage.

1 Introduction

Poker folklore holds that players leak the strength of their hands through nonverbal "tells," and the 2026 World Series of Poker broadcast popularized the idea that such leakage is machine-readable, overlaying a behavioral hand-strength model on live coverage. The scientific evidence behind that idea is thinner than its television presence suggests. The best controlled result reports that naive observers judge professional players' hand quality above chance from arm movements alone, at modest effect sizes, while judgments from the face are at or below chance (Slepian et al., 2013). Automated deception detection benchmarks report within-domain accuracies in the sixties and cross-domain transfer barely above chance (Guo et al., 2023). No prior work, to our knowledge, has tested the broadcast-poker claim end to end: does adding automatically extracted nonverbal behavior improve prediction of hand strength over the betting action alone, evaluated out of session, on public footage?

This paper contributes four things.

(1) A reproducible pipeline that converts raw broadcast video into a labeled decision-level dataset: OCR of the broadcast graphics package into per-second game-state snapshots, a state machine that reassembles hands and decision windows, Monte Carlo equity labels from the exposed hole cards, and per-decision behavioral features from face landmarks, blendshapes, and body pose, with per-player normalization and explicit bet-size leakage controls.

(2) An identity-contamination case study (Section 4). Behavioral datasets built on multi-person footage must attribute every measured face to the right person. We document how three small engineering gaps (an open-set similarity threshold, nondeterministic reference enrollment, and unbounded nearest-position matching) let a lookalike neighbor's face be silently measured as the tracked player's, and how the resulting artifact produced an apparently resolvable behavioral effect that grew under partial cleaning and vanished under full cleaning (Figure 1). We distill the fix into a verification protocol.

(3) A preregistered, adequately powered replication failure (Section 5). With the analysis frozen before the new footage was downloaded, face and pose features reduce pooled out-of-session AUC by 0.075 (95% CI $[-0.134, -0.020]$). The study had approximately 91% power to detect the registered positive estimate.

(4) An anti-transfer finding (Section 6). Behavioral features are not merely uninformative: a behavior-only model scores AUC 0.349 out of session, meaning learned behavior-to-strength associations systematically invert across sessions, including within the same player. Session-level normalization removes only part of the inversion. This is consistent with tells that are contextual and unstable rather than absent, but either way they are unusable by models of this class at this scale.

2 Related work

Behavioral cues to poker hand strength. Slepian et al. (2013) showed observers rate hand quality above chance from short clips of arm motion (smoothness of the chip push), while face-only clips were directionally below chance. This motivates our smoothness features and our expectation that any true effect is small. Work on automated deception detection in the wild reports within-domain accuracy around 60 to 67 percent on acted and courtroom corpora and severe degradation across domains (Guo et al., 2023). Our result extends this pattern to a naturalistic, high-stakes, repeated-game setting with objective ground truth.

Person identification hygiene. Face verification under compression, occlusion (sunglasses are endemic at poker tables), and pose variation is well studied, but its role as a silent label-noise channel in behavioral datasets is underappreciated. Our artifact is a concrete instance of attribution noise manufacturing a positive finding, a failure mode related to identity leakage discussed in re-identification literature but, in our case, corrupting the features rather than the labels.

Methodological safeguards. We use preregistration (Nosek et al., 2018), grouped bootstrap confidence intervals (Efron and Tibshirani, 1993), and leakage review of engineered features. Trajectory smoothness metrics follow Balasubramanian et al. (2015) and Hogan and Sternad (2009); keypoint trajectories are One-Euro filtered (Casiez et al., 2012) before any derivative is taken; face and pose streams come from MediaPipe (Lugaresi et al., 2019); OCR uses PP-OCR (Du et al., 2020).

3 Data and pipeline

Footage. Four complete Hustler Casino Live cash-game sessions (February to July 2026, 22 hours total) featuring one high-volume professional (all four sessions) and a second recreational player (one session). The stream exposes hole cards in its graphics, providing ground truth without any inference from behavior.

Game state. Region-cropped OCR reads the pot, stakes, board, and per-player panels (name, stack, action text, amount to call) once per second; card sprites are template-matched with a saturation-gated segmentation robust to the two background styles in our footage. A state machine folds snapshots into hands and decision windows: a window opens when a player's panel shows an amount to call and closes at their action flip; cutaway gaps are re-merged when positions and hole cards agree; a preroll filter drops inset replays of older sessions. Across the four sessions this yields 834 hands and 5,639 decisions, 4,815 of them with equity labels (Monte Carlo equity versus a random hand, deterministic seeds; six strength classes; `is_weak` marks decisions taken with equity below 0.45, `is_bluff` marks aggressive such decisions).

Behavioral features. For each decision window of a mapped player, frames are searched for the player's face (Section 4); accepted frames contribute blendshape series (blink rate, smile asymmetry and mean, gaze dispersion, brow activity) and a pose crop that follows the accepted face contributes posture and wrist kinematics (lean variability, head motion, normalized wrist peak speed, spectral arc length, negative log dimensionless jerk, which is amplitude-invariant by construction so it cannot encode bet size). Six event detectors (downward-gaze rate, hand near face, freeze fraction and longest freeze, chip-shuffle periodicity, forward lean) are rates, fractions, ratios, or shoulder-width geometry, and passed the same leakage review; they were separately shown to carry no signal and are excluded from the primary test by preregistration. All features are z-scored per player. Coverage is honest and low: the primary player's face passes identity gating in 10 to 33 percent of window frames depending on session, and 292 labeled decisions meet the 0.5 coverage threshold.

Evaluation. Leave-one-session-out (LOSO): models train on all other sessions and predict the held-out session; predictions are pooled. Baselines use betting features only (position, sizing ratios, pot odds, street, prior raises, stack-to-pot, timing). Deltas are reported with hand-grouped bootstrap 95% CIs (2,000 resamples), resampling hands so correlated decisions within a hand never narrow the interval.

4 The identity-contamination artifact

Attributing faces on a multi-person broadcast requires deciding, per frame, whether a detected face is the tracked player. Our initial design used an open-set rule: accept the first face whose equalized grayscale patch correlates at least 0.55 with any enrolled reference patch of the player, a threshold chosen from a measured same-player/cross-player separation of 0.74 versus 0.25. Three gaps turned this into a contamination channel.

Gap 1: lookalikes above threshold. A neighboring player correlated 0.66 with the tracked player's references. Whenever the tracked player's face dipped out of detectability (chin on hand, looking down), the neighbor was accepted in his place. In one session, 82 of 193 windows contained some wrong-person frames; one window was entirely the neighbor.

Gap 2: nondeterministic enrollment. Reference patches were enrolled by seeking to hand-verified timestamps and taking the detected face nearest a specified position. Detection under a warm video-mode tracker is stateful: which faces are found at a seek depends on the seeks before

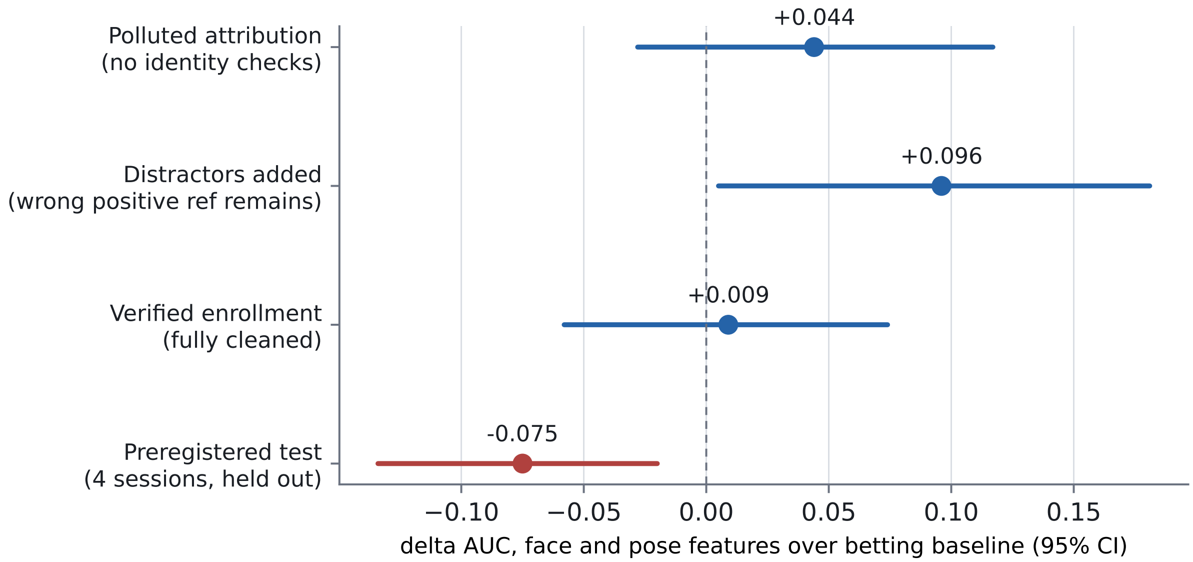

Figure 1: An apparent behavioral effect tracks attribution pollution, not behavior. Points are ΔAUC of face and pose features over the betting baseline for `is_weak` with hand-grouped bootstrap 95% CIs. The first three estimates are in-sample over two sessions under progressively stricter identity hygiene; the fourth is the preregistered four-session test.

it. Re-runs enrolled different faces from the same specification.

Gap 3: unbounded nearest-position matching. When the intended player went undetected at a reference timestamp, the nearest detected face was enrolled with no distance bound. In two of four sessions this placed a neighbor's face inside the player's positive reference library, which also neutralized the distractor defense added after Gap 1: the enrolled lookalike outscored his own distractor reference.

Figure 1 shows the consequence. With no identity checks the apparent effect was $+0.044$ $[-0.028, +0.117]$. After registering the known lookalike as a distractor, but with a wrong face still enrolled as a positive reference, the effect strengthened to $+0.096$ $[+0.005, +0.181]$, nominally resolvable. With enrollment made deterministic (fresh tracker per reference frame), position-bounded (a face must lie within 110 px of its specification), and every enrolled patch visually verified, the in-sample effect collapsed to $+0.009$ $[-0.058, +0.074]$. The mechanism is mundane: another person's face, measured during the tracked player's decisions, covaries with decision context (cameras cut to neighbors during long tanks) in ways that imitate a tell.

Protocol. The resulting checklist is short: (i) enroll references deterministically, one detector instance per reference frame; (ii) bound enrollment by position and refuse specifications whose intended face is not found; (iii) verify every enrolled patch by eye; (iv) sweep each session for faces that score above threshold against the references, and register every non-target such face as a distractor; (v) attribute a face only if it clears the threshold and beats every distractor score by a margin, best candidate wins; (vi) treat staff and commentators as candidate distractors too (one of ours scored 0.60).

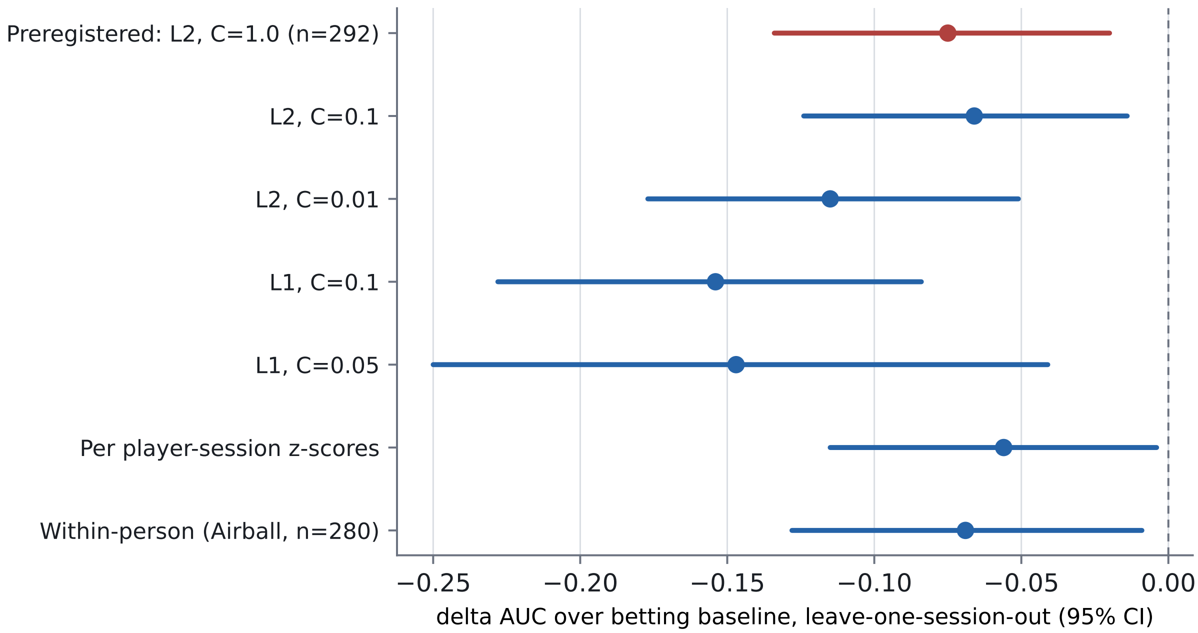

Figure 2: No analysis choice rescues the features. All arms are ΔAUC over the betting baseline for `is_weak`, pooled LOSO, hand-grouped bootstrap 95% CIs.

5 Preregistered replication

Before downloading any new footage we froze, in a committed preregistration document: the hypothesis (face and pose features improve pooled LOSO AUC for `is_weak`), the registered estimate (+0.096), the exact pipeline settings, the seat-mapping discipline for new sessions (commit-moment identification and a distractor audit before any model output), the pass criterion (pooled CI excluding zero on the positive side), and a prohibition on post-hoc adjustment. Two sessions were selected by roster and layout compatibility; one candidate was excluded before extraction because it used a different graphics package, and the exclusion was logged in the preregistration.

The pooled four-session test returned

$$\Delta\text{AUC} = -0.075, \qquad 95\%\ \text{CI}\ [-0.134,\ -0.020], \qquad n = 292.$$

The CI excludes zero on the negative side: added behavioral features reduce out-of-session performance. Under a normal approximation to the bootstrap distribution (SE $\approx$ 0.029), the test had approximately 91% power to detect the registered +0.096 and 80% power for any delta of at least +0.082; the upper confidence bound rules out all positive effects at the 2.5% level. The betting-only baseline itself scores AUC 0.549 pooled out of session, against 0.75 within session, replicating the known brutality of cross-session transfer.

6 Why negative, not just null

A significantly negative delta invites the objection that the estimator, not the signal, is at fault: ten added dimensions can hurt a small model even when they carry information. Four analyses reject this reading (Figure 2).

Regularization does not help. Stronger L2 shrinkage leaves the delta unchanged or worse ($C = 0.01$: -0.115), and sparse L1 selection is worst of all (-0.154): selecting the few behavioral features that look best in training amplifies reliance on associations that do not hold in the held-out session.

Behavior alone is anti-predictive. A behavior-only model scores AUC 0.349, far below chance. Whatever association the model learns between behavior and weakness in three sessions, the reverse tends to hold in the fourth. Under label exchange this would be signal; under session exchange it is instability.

The inversion survives within a person. Restricted to the primary player's four sessions (one identity, one seat style, $n = 280$), the delta is -0.069 $[-0.128, -0.009]$. Idiosyncrasy across people does not explain the failure; instability within a person across nights does.

Session geometry explains only part. Re-standardizing features per player-session (removing session-level offsets induced by camera geometry, since a lean angle or apparent motion scale depends on the shot) moves the behavior-only AUC from 0.349 to 0.387 and the delta to -0.056 $[-0.115, -0.004]$. Session offsets are therefore a real confound channel that future work must normalize away, and still not the whole story.

7 Discussion

Three readings of these results deserve separation. The narrow reading is engineering: with tabular summary features over MediaPipe streams, a linear model, four sessions, and camera-limited coverage, behavioral signal is not extractable, and the attempt costs accuracy. The middle reading is statistical: effects of the size the literature supports (a few hundredths of AUC) require an order of magnitude more covered decisions than 22 hours of broadcast yields, and any study in this regime that reports a large positive behavioral effect should be audited first for identity contamination, which we show can manufacture exactly such an effect. The broad reading is substantive: even within one professional, behavior-strength associations invert across nights, which is what one expects if experienced players' surface behavior is dominated by context, table dynamics, and deliberate randomization rather than stable leakage. Our data cannot distinguish "no tell exists" from "tells exist but are contextual and unstable"; both entail that broadcast-scale models of this class cannot use them.

The artifact deserves the emphasis we give it. It produced, in our own hands, a preregistration-worthy positive estimate with a confidence interval excluding zero, out of nothing but face misattribution, and it grew stronger under partial cleaning because the partially cleaned pipeline retained the most correlated contamination. Behavioral-video findings built on multi-person footage without an explicit attribution audit should be treated as provisional.

8 Limitations

One primary player carries most covered decisions; sunglasses make eye features noise for him, as for many poker subjects. Coverage is bound by broadcast camera direction, not by us. Card-reading errors survive at a low rate (rank confusions on small sprites) and add label noise to equity. The betting baseline is deliberately simple; a stronger baseline would only shrink headroom for behavior. Our features are window-level summaries; sequence models over raw streams remain untested here, though they face the same coverage and identity constraints with more capacity to overfit them.

9 Ethics and reproducibility

All footage is publicly broadcast with player consent to the stream's hole-card exposure; we analyze it post hoc, redistribute no footage or identifiable imagery, and publish code, derived features, timestamps, and the preregistration rather than clips. Pipeline, analysis code, seat maps, audit tooling, and the frozen preregistration with its logged result are available in the project repository.

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